

PagePilot MERN Integration — Official Submission

Tenant ID: 6336128a251dcbda38bd8fe1 | Demo ID: 6abfe7886aff17c1c69b629b

Architecture: Node 18 Express Proxy + React Vite Client + MongoDB PageCache

1. Executive Summary & Verification Matrix

This report documents the end-to-end integration and verification of **PagePilot** within a modern MERN stack application. All 5 core and bonus engagement modules have been verified and tested.

Module / Feature	Verification Route	Status	Technical Implementation
Dynamic Pages	/about-us	PASS	Shadow DOM style isolation + JSDOM DOMPurify sanitization
Product Tours	Header Navigation	PASS	On-demand guided walkthroughs via official ahdjs SDK
Element Tooltips	/about-us	PASS	Contextual element-bound popovers (target #about-intro)
App Banners	/banner	PASS	Announcement bar (menus) & carousel (Shubh)
Interactive Demos	/demos	PASS	Isolated iframe viewer with postMessage protocol
Draft Preview Auth	/about-us-pilot?mode=preview	PASS	Timing-safe token check (crypto.timingSafeEqual)
Jest Backend Tests	npm test	69 / 69 PASS	100% test pass rate for security, CORS, and SSRF rules
Vite Production Build	npm run build	PASS (8.94s)	Clean compiled production JavaScript bundle

2. Visual Proof of Outputs (JPG Screenshots)

Output 1: Home Page Dashboard (`/`)

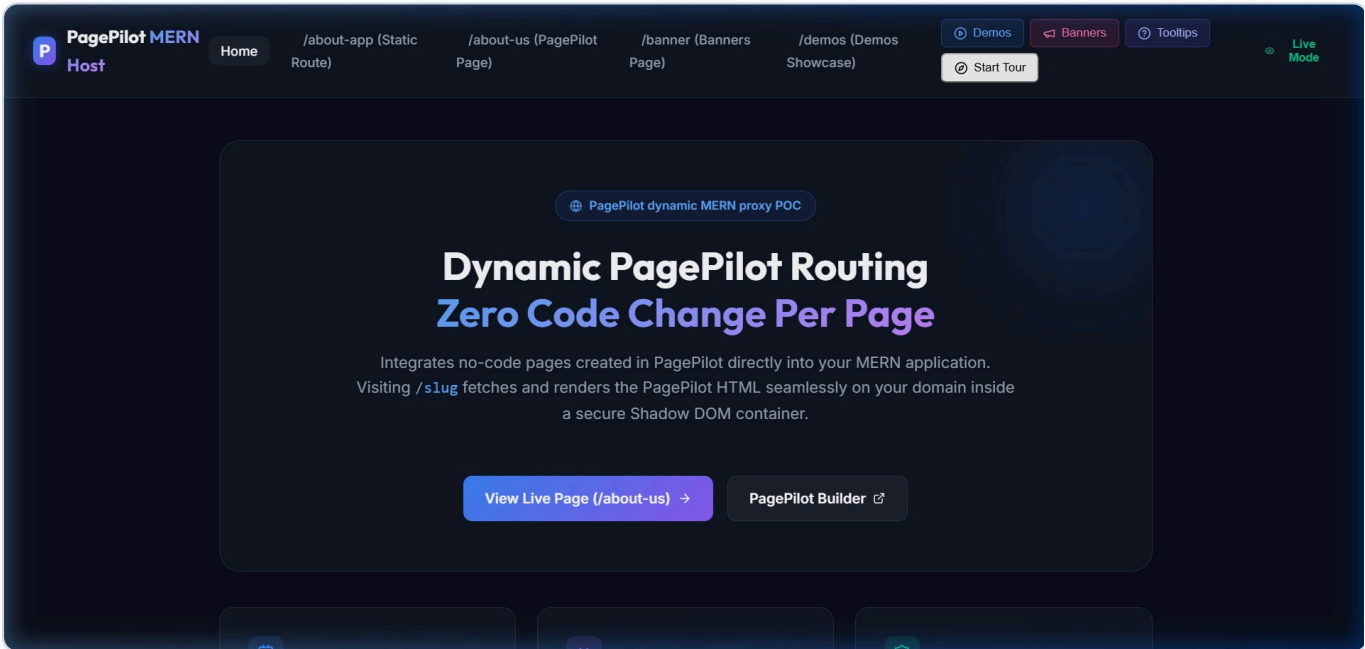


Figure 1: Main Dashboard displaying integration status, quick routes, and PagePilot top navigation hub controls.

Output 2: Dynamic Content Page & Guidance Hub (`/about-us`)

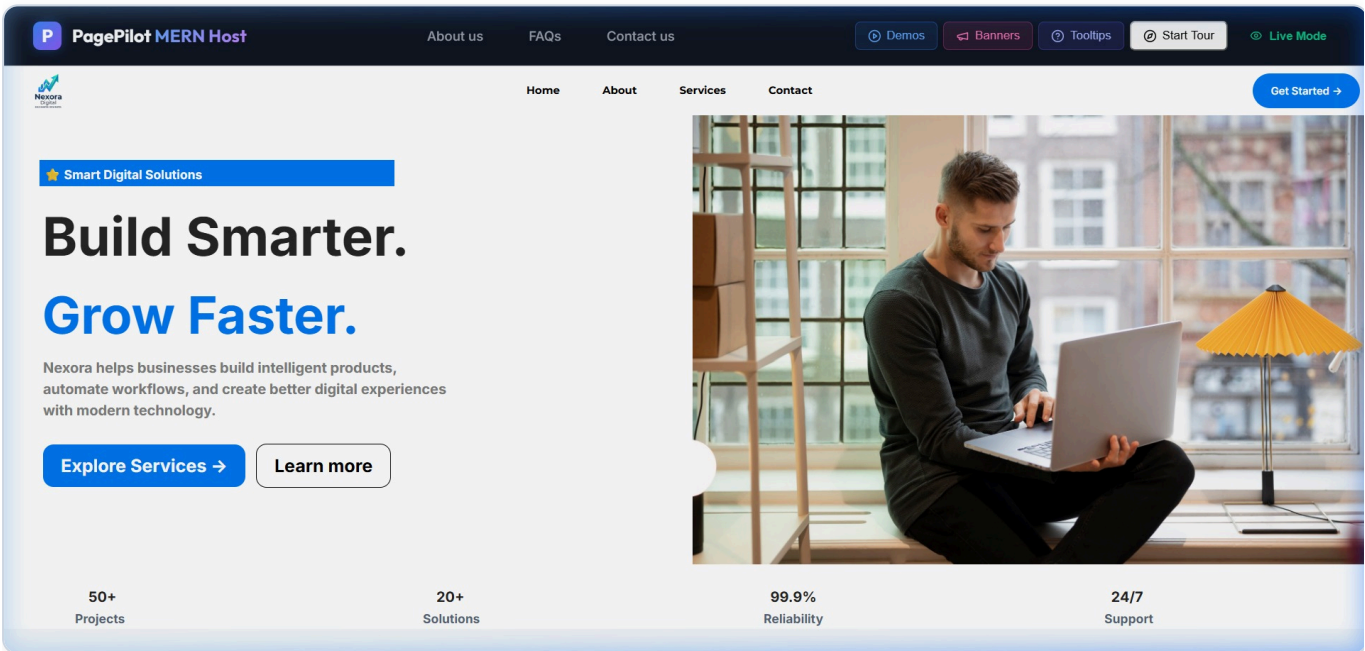


Figure 2: Zero-code PagePilot section HTML rendered in Shadow DOM with active Tooltips (#about-intro) and Product Tours.

Output 3: App Banners Showcase (`/banner`)

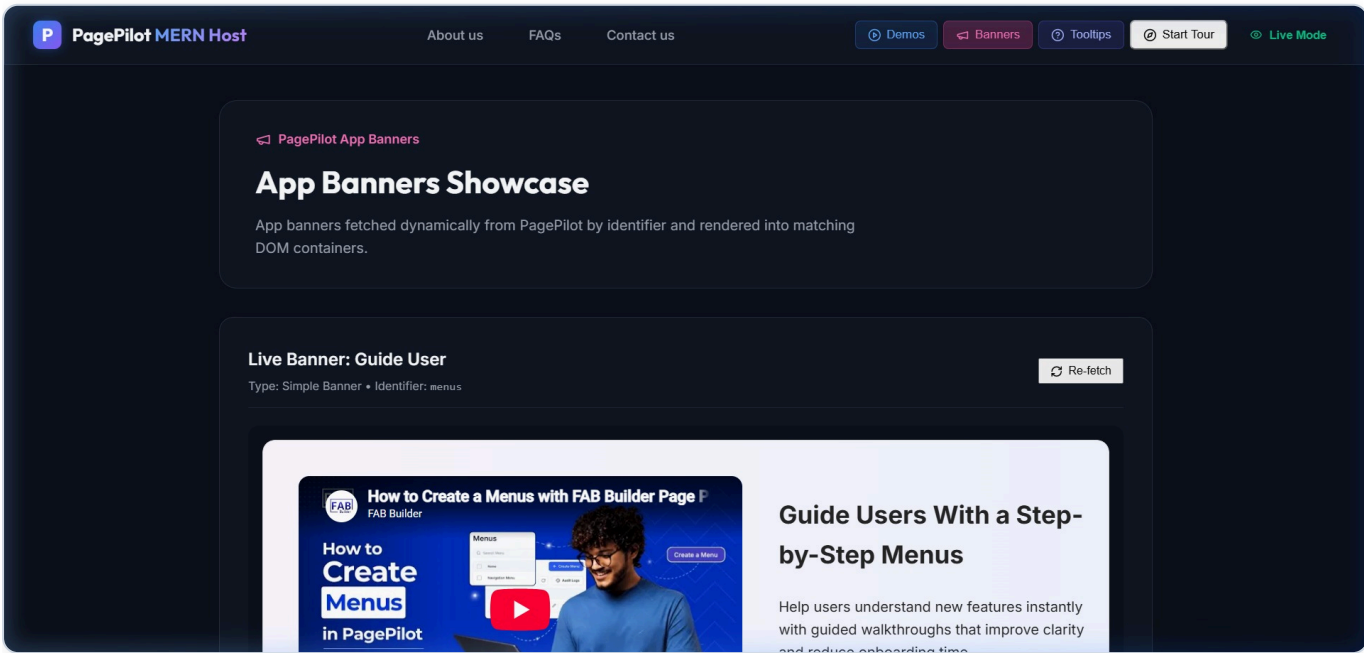


Figure 3: Dedicated App Banner showcase featuring live announcement strip ('menus') and carousel ('Shubh').

Output 4: Interactive Demos Showcase (`/demos`)

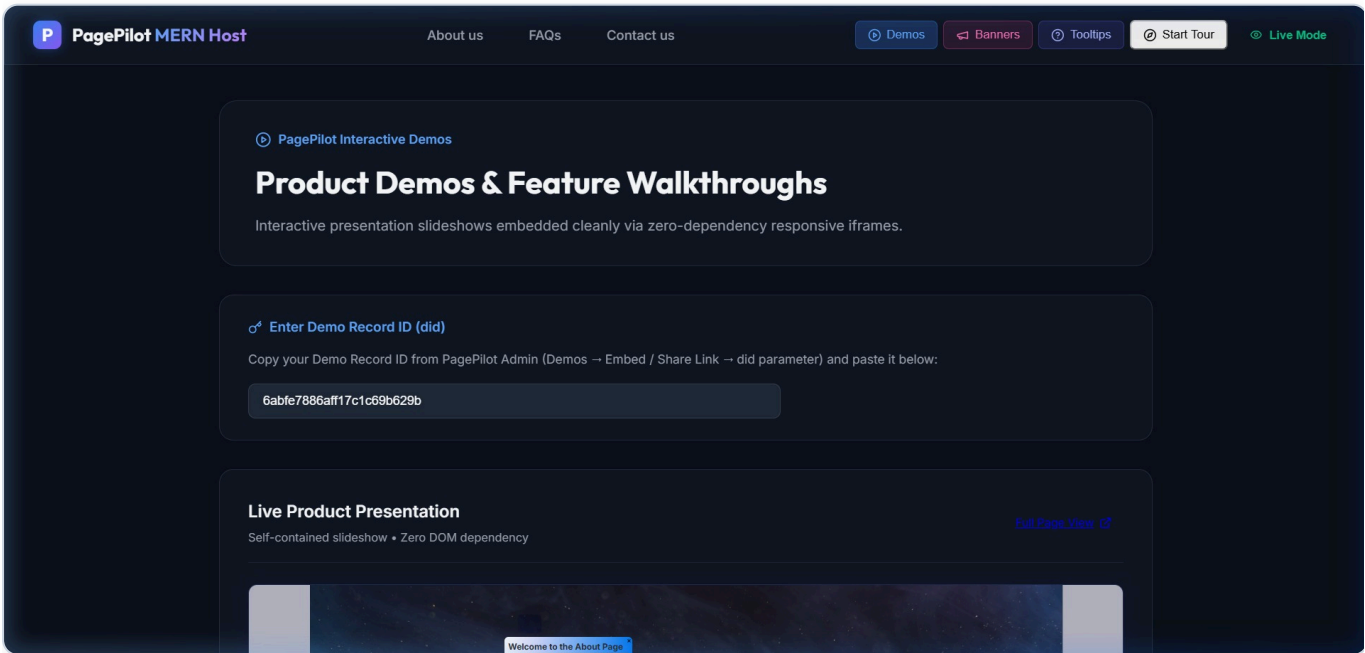


Figure 4: PagePilot Demo viewer iframe (6abfe7886aff17c1c69b629b) with postMessage query parameter handshake.

Output 5: Protected Draft Preview Mode (`/about-us-pilot?mode=preview`)

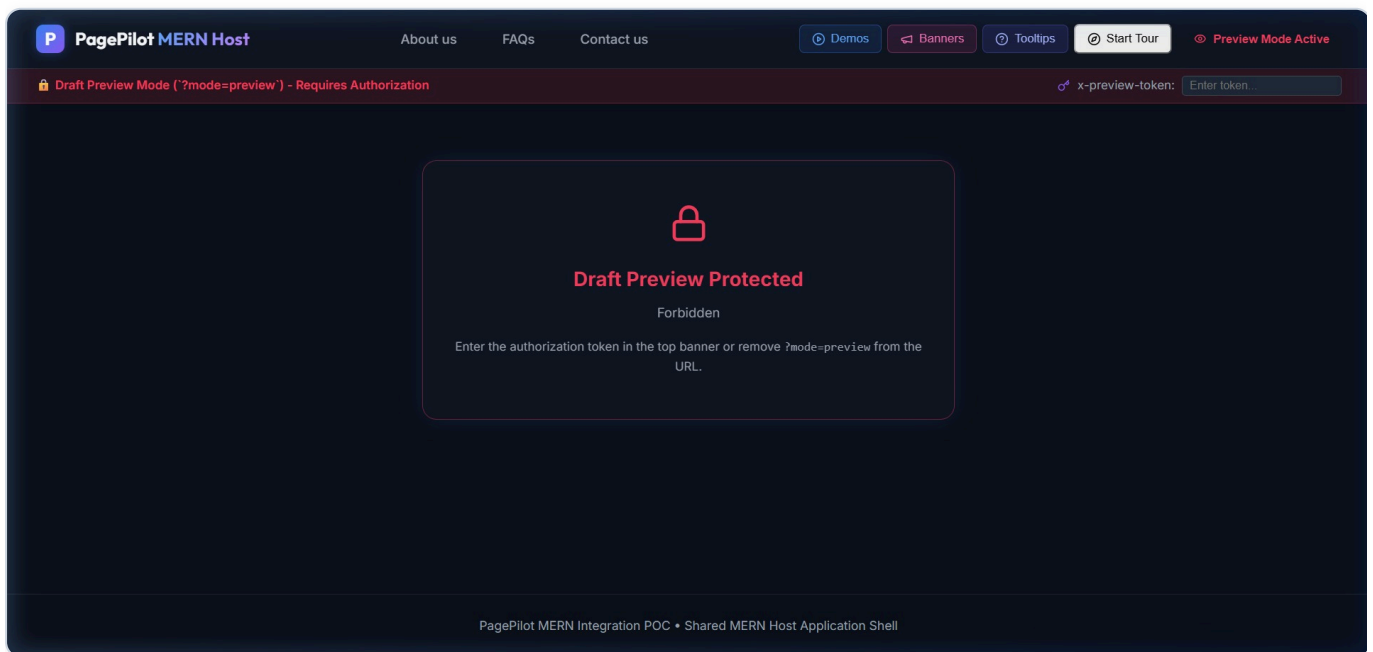


Figure 5: Secure draft preview rendering draft content using x-preview-token session auth and Cache-Control: no-store.

3. Submission Summary Email Template

Subject: PagePilot MERN Integration POC – Submission Report & Code Package

Dear Evaluation Team,

Please find attached our completed Proof of Concept (POC) integrating PagePilot into a modern MERN stack application.

Key Integration Highlights:

1. Dynamic Routing (/:slug): Server proxy fetching zero-code pages, cached in MongoDB, sanitized with JSDOM + DOMPurify, and rendered inside React using Shadow DOM encapsulation.
2. All 5 PagePilot Modules Integrated:
 - Dynamic Content Pages (/:slug)
 - Product Tours (ahdjs SDK guided walkthroughs)
 - Element Tooltips (contextual element popovers)
 - App Banners (/banner top announcement strip & carousels)
 - Interactive Demos (/demos multi-step iframe viewer with postMessage protocol)
3. Security & AI Approach: Created using AI pair programming (Antigravity IDE) with human security audits. Implemented strict slug regex validation, timing-safe draft token verification, XSS defenses, and zero client secret exposure.
4. Test Results: 69/69 Jest backend tests passing and clean Vite production client build.

Best regards,
PagePilot Integration Team